

Map passes: speedup over Vanilla Gibbon, per program

outlined: beyond the 32×/0.5× colour range ▼: slower than the column to its left corner: same configuration, auto-vectorizer on †: auto-vectorizer off in Vanilla Gibbon too

Real-world	ColorOctree	1×	1.01× 1.01×	▼ 0.581× 0.602×	0.746× 0.758×	1.83× 1.84×	1.94× 1.96×	1.94× 1.96×
	Compiler	1×	2.6× 2.63×	▼ 2.07× 2.22×	2.07× 2.08×	20.1× 20.7×	20.1× 20.7×	20.1× 20.7×
	DBQuery	1×	1.17× 1.18×	▼ 0.946× 0.913×	1.03× 1.02×	6.21× 6.05×	▼ 6.2× 6.21×	6.2× 6.21×
	DomTree	1×	1.01× 1.04×	▼ 0.724× 0.745×	0.754× 0.771×	1.05× 1.09×	1.05× 1.09×	1.05× 1.09×
	ObjectGraph	1×	1.13× 1.14×	▼ 1.05× 0.873×	1.4× 1.52×	2.32× 2.45×	2.35× 2.43×	2.35× 2.43×
	OctTree	1×	▼ 0.973× 0.995×	▼ 0.817× 0.823×	1.04× 1.09×	5.89× 5.87×	5.89× 5.88×	5.89× 5.88×
	PiecewiseFunctions	1×	1.04× 1.05×	▼ 0.936× 0.959×	1.25× 1.3×	5.15× 5.32×	5.29× 5.31×	5.29× 5.31×
	Trie	1×	1.03× 1.05×	▼ 0.903× 0.894×	▼ 0.894× 0.901×	3.88× 3.88×	3.88× 3.93×	3.88× 3.93×
Synthetic	Add1TreeInt8	1×	1.2× 1.12×	1.26× 1.28×	4.37× 6.67×	12.5× 22.6×	22.7× 22.5×	22.7× 22.5×
	Add1TreeInt16	1×	1.38× 1.35×	▼ 1.3× 1.31×	4.01× 5.52×	8.87× 11.8×	11.7× 11.7×	11.7× 11.7×
	Add1TreeInt32	1×	1.29× 1.19×	▼ 1.24× 1.24×	3.02× 3.68×	5.21× 5.8×	5.8× 5.8×	5.8× 5.8×
	Add1TreeInt64	1×	1.18× 1.24×	1.19× 1.2×	2.23× 2.5×	3.15× 3.27×	3.27× 3.26×	3.27× 3.26×
	LinearListReduction	1×	2.11× 2.13×	▼ 1.64× 1.77×	1.67× 1.67×	4.64× 4.76×	5.27× 5.3×	5.27× 5.3×
	MonoTree	1×	1.32× 1.3×	▼ 1.22× 1.22×	2.37× 2.65×	3.35× 3.48×	3.48× 3.48×	3.48× 3.48×
	TernaryTree	1×	1.06× 1.06×	1.06× 1.05×	1.58× 1.71×	1.9× 1.97×	1.97× 1.97×	1.97× 1.97×
Arithmetic intensity	ArithmeticIntensityInt8 †	1× 1.23×	1.01× 1.31×	1.02× 1.29×	1.1× 5.99×	1.14× 6.93×	6.68× 6.74×	6.68× 6.74×
	ArithmeticIntensityInt16 †	1× 2.43×	1.02× 2.74×	▼ 1.01× 2.74×	1.09× 7.61×	1.13× 9.25×	9.17× 9.22×	9.17× 9.22×
	ArithmeticIntensityInt32 †	1× 1.4×	1× 1.51×	1.01× 1.5×	1.08× 3.27×	1.13× 3.53×	3.61× 3.66×	3.61× 3.66×
	ArithmeticIntensityInt64 †	1× 0.951×	1.02× 1.02×	1.02× 1.02×	1.03× 1.04×	1.07× 1.07×	1.07× 1.08×	1.07× 1.08×
		Vanilla Gibbon (AoS)	Mutable cursors (AoS)	SoA layout	Loopification	Buffer sharing	Gibbon SIMD	Total